KSJ RESEARCH

Research Note Vol.4

Selling Power to a Data Centre

Equity in a hypothetical 35 MW solar asset — KEPCO contract (CfD), direct PPA with a data centre, or merchant: which pays for its risk?

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The data centre saves 169.8 (energy rate + fuel adjustment + climate charge + fund levy) − required PPA 152.8–164.2 = room 5.6~17.0 — at most ~17 < ~40 network charges in a public example (B)

Price evidence (official first) → cash flows and debt for three routes → break-evens (price, termination, network charges) → Monte Carlo → IC memo

Key metrics [*1,*2,*3,*4,*5]

Key metric	Value	p.
Hypothetical solar · CAPEX (backed out of LCOE, judgement)	35MW · KRW 61.1bn	1
Yr-1 output · share of data-centre use	49.1 GWh · 15%	1
Corridor: KEPCO contract ~ solar-hour saving	**147.7 ~ 165.3**	3
Ceiling: energy / +fuel adj. / +climate / +fund levy	151.3 / 156.3 / 165.3 / 169.8	3
Saving · spring-autumn / summer / winter	140.9 / 181.0 / 186.6	5
Equity IRR · CfD / PPA (mid) / merchant	**12.2% / 13.3% / 3.7%**	4
Gearing · CfD / PPA / merchant	80% / 75% / 38%	4
Required PPA premium (1% term., 0–2pp spread, NPV)	**+5.1 ~ +16.5**	5
Buyer's room for network charges (saving + fund levy)	**5.6~17.0**	5
Public example of charges (KEPCO-brokered 3rd-party PPA, solar Case 1, B)	**~KRW 40**	3
Monte Carlo · PPA equity IRR P5 / P50	6.4% / 13.0%	7
Monte Carlo · merchant loss probability (IRR < 0)	5%	7

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *4 SFOC, 'Stranded RE100' (Dec 2021), printed p.9 (PDF p.11) table 2 — KEPCO's third-party PPA charge simulator *5 KSJ Research
Note: KSJ Research. Asset and offtaker are hypothetical. Prices KRW/kWh, amounts KRW 100mn. Source grades and provisional inputs in A2–A4

13.3%

Equity IRR · direct PPA (corridor mid KRW 156.5/kWh, no termination)

Same plant, different buyer: equity IRR Merchant 3.7% → KEPCO CfD 12.2% → Direct PPA 13.3%

- KEPCO: the 2026 auction cap KRW 147.7/kWh as a proxy for the 2027 contract market → gearing 80%, IRR 12.2%
- Merchant: SMP KRW 113.2/kWh (12-month mainland average), no RECs for post-2027 entrants → 3.7%
- IRR levels depend on CAPEX (backed out of LCOE, judgement) — at 2022 actual cost KEPCO IRR is 23.7%; the price gap between routes barely moves

+5~16

Required PPA premium (1% termination, PPA discount +0–2pp, per kWh)

Price to accept a PPA instead of KEPCO: KRW 152.8/kWh–KRW 164.2/kWh

- Rule (NPV): PPA price at which expected equity NPV equals KEPCO's — PPA flows discounted at the hurdle plus a 0–2pp risk spread; SMP after termination
- Judgement that moves it: PPA gearing 65% → +9, 80% → +3 (spread 0)
- Termination rate and spread have no public basis, so grids are shown; a deal sets them from the offtaker's credit

6~17

Buyer's room for network and other charges (solar-hour saving + fund levy 169.8, per kWh)

Saving 169.8 − required PPA = room 5.6~17.0 < ~40 of charges in a public example

- Buyer ceiling in 4 stages: energy rate 151.3 → +fuel adj. 156.3 → +climate 165.3 → +fund levy 169.8 — the PPA price ceiling is 165.3 (the levy also hits the PPA payment and cancels); room is measured with the levy. Demand charges are not avoided. On the 181.9 average tariff the room would be 23~34
- Seasons split: spring/autumn 140.9 · summer 181.0 · winter 186.6 — spring/autumn below the KEPCO price 147.7
- To close, the buyer must pay extra (RE100 value) = required + charges − saving → 0–9 with the 9.73 KPX settlement alone, 23–34 at the 40 example

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PPA equity IRR P50 13.0% · P5 6.4% — below KEPCO 12% of the time; merchant loses money 5% of the time

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The buyer's room of at most ~17 falls short of the ~40 public network-charge example — confirm RE100 willingness to pay and actual charges first, then termination protection

App. — **Appendix** — A1–A5
A1 IC memo · A2–A3 Inputs and source grades · A4 Data notes, method · A5 Glossary

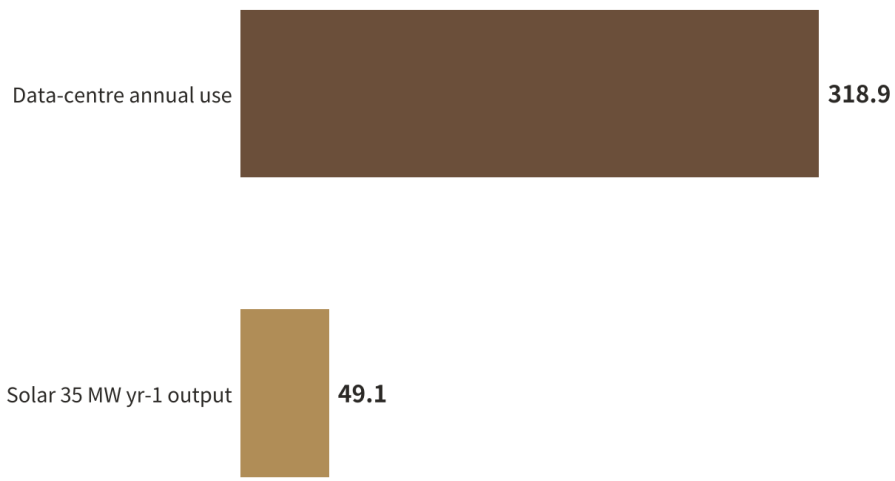
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Starting point: a 40 MW data centre's power bill is KRW 58.0bn a year — 35 MW of solar sized within its load covers 15%

Vol.2 data centre (lender's view) → Vol.4 the plant that sells it power (equity investor's view)

Data centre 318.9 GWh a year vs solar 49.1 GWh — 15% supplied directly [*1,*2]

Unit: GWh



Source: *1 KSJ Research Vol.2 (Oct 2026) *2 KSJ Research
Note: Vol.2 Korean case (IT 40 MW). 35 MW of solar = peak output within the data centre's 36.4 MW average load, so all output is consumed; nights and gaps come from KEPCO

Power bill KRW 58.0bn a year — the cost Vol.2 passed to tenants [*1,*2,*3,*4]

Step	Value
IT capacity × load × PUE	40MW × 70% × 1.3
× 8,760 h = annual use	318.9 GWh
× KEPCO industrial average tariff (2025)	KRW 181.9/kWh
= Annual power bill	**KRW 58.0bn**
Solar MW × capacity factor × 8,760 h	35MW × 16%
= Yr-1 output (0.5% annual degradation)	**49.1 GWh**

Source: *1 KSJ Research Vol.2 (Oct 2026) *2 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *3 NREL Jordan & Kurtz (2012), PV degradation review *4 KSJ Research
Note: For the annual bill = average tariff incl. demand charges. A PPA saves less — the solar-hour saving (energy rate + fuel adjustment + climate charge), KRW 165.3/kWh (p3, p5)

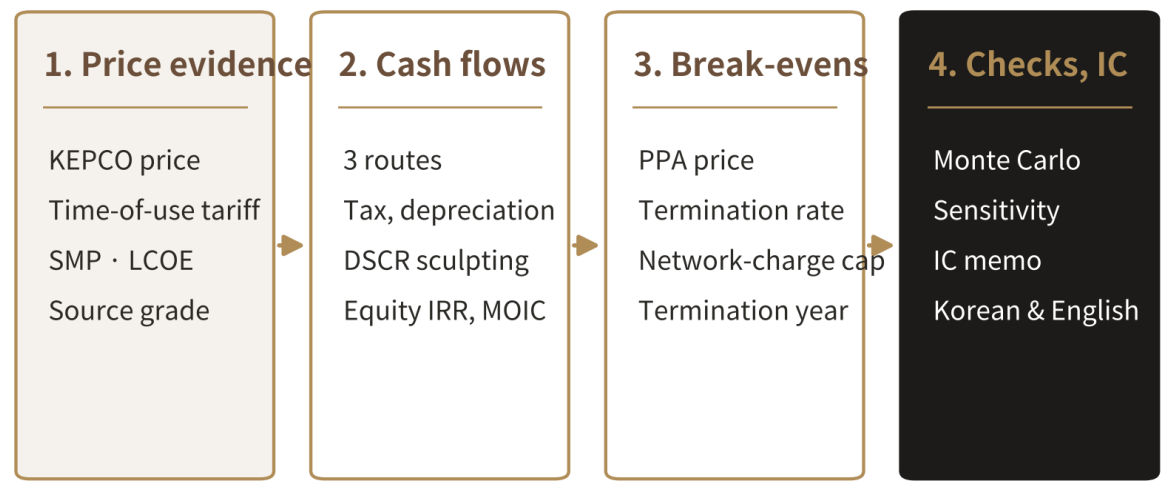
Question: if the plant sells to the data centre instead of KEPCO, does the extra price pay for the risk? [*1,*2]

An equity investor's three routes

- ① KEPCO contract (CfD): standard for new solar from 2027; 20-yr fixed price, KEPCO as counterparty
- ② Direct PPA: negotiated price; counterparty is Vol.2's data-centre SPV
- ③ Merchant (SMP): full price risk; no RECs for new entrants
- Vol.2 took the lender's view, Vol.3 the ECM view — this note takes the equity investor's

Source: *1 KSJ Research Vol.2 (Oct 2026) *2 KSJ Research

Structure: prices → cash flows → break-evens → checks and IC memo [*1]



Source: *1 KSJ Research

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Rules: from 2027 new solar defaults to a 20-yr KEPCO contract — a PPA is the alternative, merchant sales get no RECs

Contract market · REC sunset · distributed-energy cap · AI data-centre act → three routes

Timeline: from 2027 a KEPCO long-term contract is the default, a direct PPA the alternative [*1,*2,*3,*4]

Date	Change	Grade
2026-02-03	Distributed-energy zones: firms on local grids to be allowed renewable PPAs; 35 MW cap under review	Press report / secondary report
2026-05-07	AI Data Centre Act passed — the requested LNG direct-PPA carve-out was dropped	Press report / secondary report
2026-05	Distributed resources capped at 40 MW in principle; a data-centre carve-out bill is pending (ministry reported opposed)	Press report / secondary report
2026-07-16	2026 round-1 solar auction: cap KRW 147.686/kWh (about 5% lower), ~1,000 MW	Official source
2026-09-15	Renewable Energy Act amended (Act No. 21943) — a contract market incl. long-term fixed-price contracts (art. 12-14); a single KEPCO channel is per press reports	Official source
2027-01-01	Effective — REC article (12-7) in force to 31 Dec 2029, issuance deadline 31 Dec 2026 · existing holders and pre-act licensees recognised by notice keep the former rules (addendum arts. 2–3)	Official source

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Renewable Energy Act amendment (Act No. 21943, promulgated 15 Sep 2026), art. 12-14 and addendum arts. 1–4 — law.go.kr original *3 Financial News (15 Sep 2026, citing the ministry): Renewable Energy Act reform *4 Herald K (9 May 2026) · Hankyung (3 Feb 2026): distributed energy, AI data-centre act
Note: Grades: official = government release checked · press/outside report = media citing a government release (original not yet matched) — dates and content as reported

Three routes: price, counterparty and debt terms move together [*1,*2,*3]

Item	KEPCO contract (CfD)	Direct PPA with data centre	Merchant (SMP)
Price	Auction price fixed 20 yrs (cap 147.7)	Negotiated between KEPCO price and buyer's saving	SMP (no RECs)
Counterparty	KEPCO	Data-centre SPV (Vol.2)	Market
Term	20 yrs	20 yrs (U.S. nuclear PPA precedents)	—
Target DSCR · max gearing	1.20 · 80%	1.30 · 75%	1.60 · 55%
Tenor	18 yrs	18 yrs	12 yrs
Equity risk	Capped upside · losing the auction	Offtaker credit · network charges · termination	All price risk

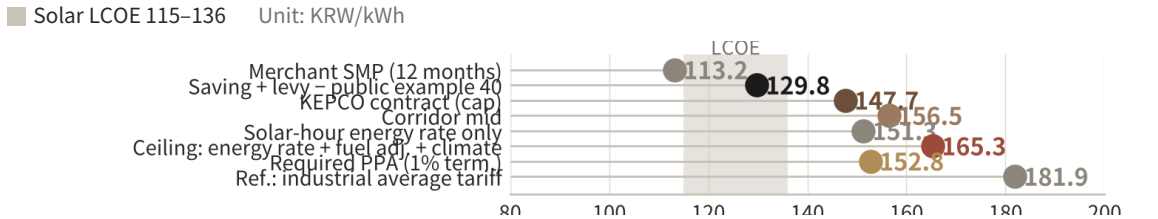
Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Financial News (15 Sep 2026, citing the ministry): Renewable Energy Act reform *3 KSJ Research
Note: Debt terms are author judgments (no public term sheets): weaker counterparty → higher DSCR, lower gearing. Rate = KTB 3-yr 3.84% (2026-10-20) + 150bp

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Price corridor: KEPCO 147.7 to the solar-hour saving 165.3 — 16.6 narrower than on the average tariff (181.9)

Floor = seller's alternative · ceiling = buyer's alternative (solar-hour rate) · cost = LCOE · merchant = SMP · U.S. check

Corridor 147.7–165.3 — net of ~40 in public-example charges the buyer's ceiling falls to 129.8 (levy-matched) and the corridor closes *1,*2,*3,*4,*5,*6,*7,*8,*9,*10,*11



Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KEPCO Monthly Statistics no.566, table 16-1 (2025 monthly solar output) *4 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *5 KEPCO ON cost-linked tariff (fuel cost adjustment) — annual band on the adjustment rate *6 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *7 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *8 SFOC, 'Stranded RE100' (Dec 2021), printed p.9 (PDF p.11) table 2 — KEPCO's third-party PPA charge simulator *9 Direct-PPA notice (MCEE 2025-61), art. 25(3), 9(4) *10 Electricity Business Act decree art. 36 *11 KSJ Research
Note: KEPCO price = 2026 round-1 cap (proxy for 2027). Buyer ceiling = energy rate 151.3 → +fuel adj. 156.3 → +climate 165.3 → +fund levy 169.8 — energy = post-reform industrial (B) HV-A option II rates in solar hours, weighted by 2025 monthly output (no demand charge; daily shape is a judgement); fuel adjustment = Q4 2026 rate (press report, B); climate charge 9.0 (KEPCO ON, effective 2023-01, official). Average tariff = reference only. LCOE = 2024 Korean range. The +5 fuel adjustment is the KEPCO rule's annual cap (±5, KEPCO ON, A) — holding it for 20 years puts the buyer ceiling at its highest, and it still does not cover network charges (room at most 17.0 < 40 example). Public example 40 = KEPCO-brokered third-party PPA, mid/large solar Case 1 (B; charge date not stated (KEPCO simulator, report published Dec 2021; only the 136 generation cost is 2020-based)) — SFOC printed p.9 (PDF p.11), table 2; breakdown in A4. A direct-PPA buyer pays T&D use charges, trading fee, settlement charges, power-industry fund levy (direct-PPA notice art. 25(3) (MCEE notice 2025-61, A)) — no fuel adjustment or climate charge → the ceiling with the 14.0 added is the like-for-like comparison. Room is measured with the 2.7% fund levy, 169.8 — details in A4. · Provisional: Daily output shape exponent, Saturday time-band rule, Fuel cost adjustment (Q4 2026) pending official source (A3)

Price evidence: 9 items — official 5 · judgement 2 · press/outside report 2 *1,*2,*3,*4,*5,*6,*7

Item (KRW/kWh)	Value (range)	Grade	Published
KEPCO contract price (2026 round-1 cap)	147.7	Author judgement	2026-07-16
Buyer ceiling: saving (HV-A option II, energy rate + fuel adjustment + climate charge)	165.3 (163.8~166.0)	Author judgement	2026-03-13
Fuel cost adjustment (Q4 2026)	5.0	Press report / secondary report	2026-09-21
Climate & environment charge	9.0	Official source	2023-01
KEPCO industrial average tariff (2025, reference)	181.9	Official source	2026-02
Solar LCOE	125.5 (115~136)	Official source	2025
SMP base	113.2 (90~148)	Official source	2026-10
Direct-PPA ancillary settlement (Jul 2023–Jun 2024)	9.7	Official source	2023-12-19
Ancillary charges, public example (KEPCO-brokered 3rd-party PPA, solar Case 1)	40.0	Press report / secondary report	2021-12

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *4 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *5 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *6 KPX Q&A (19 Dec 2023): direct-PPA ancillary settlement rate *7 SFOC, 'Stranded RE100' (Dec 2021), printed p.9 (PDF p.11) table 2 — KEPCO's third-party PPA charge simulator
Note: Saving = official tariff (13 Mar 2026) and 2025 monthly output (official) with a judgement daily shape (range = shape sensitivity) · Average tariff = 2025 industrial revenue ÷ volume (reference) · SMP = mean of monthly mainland SMP, Sep 2025–Aug 2026 (cross-checked with 10:00–16:00 hourly)

U.S. check: hyperscaler nuclear PPA at least $63/MWh — Korean solar contracts cost more *1,*2,*3

Item	USD/MWh	Grade
Talen–AWS nuclear PPA (17 yrs, 1,920 MW)	≥ 63	Official source
LevelTen N. America solar PPA avg (Q2 2026)	61.4	Press report / secondary report
Ref.: Korea KEPCO price (USD/KRW 1,332.1)	111	Author judgement
Ref.: Korea buyer's solar-hour saving	124	Author judgement

Source: *1 Talen Energy Form 8-K (11 Jun 2025, SEC EDGAR) *2 TaiyangNews (23 Jul 2026, citing LevelTen Q2 2026) *3 Bank of Korea ECOS KTB 3-yr, USD/KRW
Note: Talen: ~USD 18bn notional ÷ (1,920 MW × 8,760 h × 17 yrs) = average price at full, round-the-clock volume; the ramp to 2032 means the true price is higher (floor). Other nuclear PPAs (e.g. Meta–Constellation) are undisclosed

How to read it: the seller's floor is KEPCO's contract, the buyer's ceiling KEPCO's solar-hour rate *1

A PPA price lives between two outside options

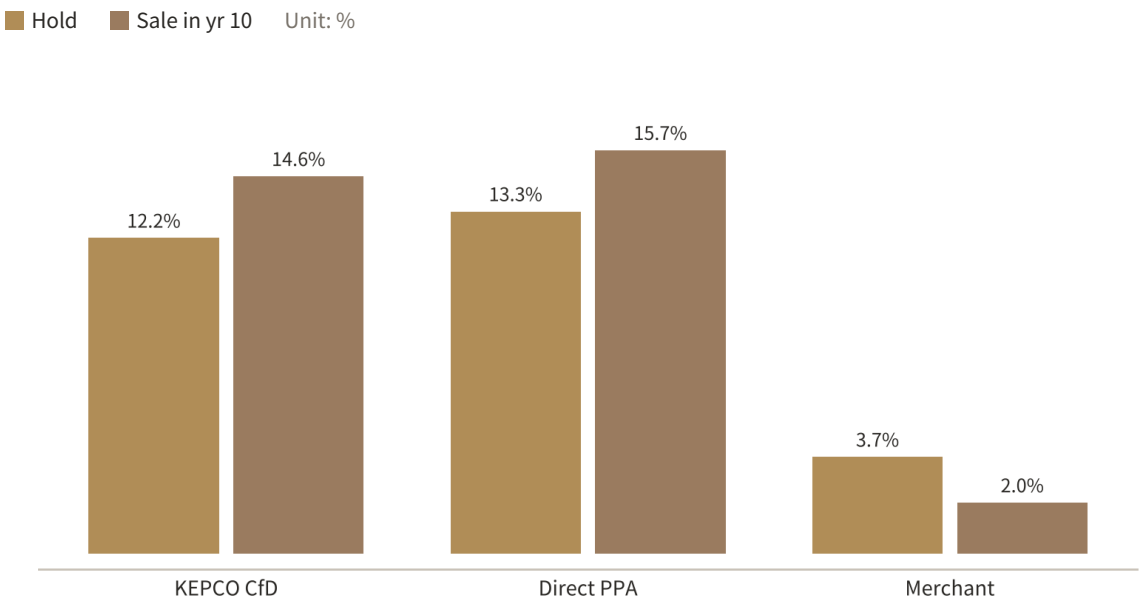
- Seller: needs more than KEPCO (147.7) for extra risk → floor + risk premium
- Buyer: PPA + charges must beat the solar-hour rate (165.3); demand charges stay. With the 40 example, the PPA must be below 129.8 (levy-matched)
- Merchant SMP (113.2) sits below the KEPCO price — no RECs for post-2027 entrants
- U.S. 24/7 nuclear at $63+/MWh — a Korean PPA is an RE100 premium product, not cheap power

Source: *1 KSJ Research

Equity IRR: PPA 13.3% · KEPCO 12.2% · merchant 3.7% — against the 8% hurdle: KEPCO CfD · Direct PPA clear, Merchant miss

Debt sizing by route → equity IRR and MOIC → exit → sensitivity

Equity IRR: hold (25 yrs) vs sale in year 10 — Direct PPA leads either way *1



Source: *1 KSJ Research
Note: PPA at corridor mid KRW 156.5/kWh, no termination. Sale price = later equity flows at the buyer's 7%

Debt sizing: the stronger the counterparty, the more debt and the less equity *1,*2,*3

Route	Debt	Gearing	Min DSCR	Equity	Eq. IRR	MOIC
KEPCO CfD	KRW 48.9bn	80%	1.33x	KRW 12.2bn	12.2%	3.96x
Direct PPA	KRW 45.8bn	75%	1.49x	KRW 15.3bn	**13.3%**	3.87x
Merchant	KRW 23.4bn	38%	1.60x	KRW 37.7bn	3.7%	1.67x

Source: *1 Bank of Korea ECOS KTB 3-yr, USD/KRW *2 Corporate Tax Act art. 55 · Local Tax Act (law.go.kr) *3 KSJ Research
Note: CAPEX KRW 61.1bn (KRW 1,746k/kW, backed out of LCOE/WACC/CF — judgement) · progressive corporate tax + 10% local · straight-line 20 yrs · debt = min(DSCR sculpting, max gearing)

Sensitivity: CAPEX and capacity factor move IRR more than SMP — contracted routes carry cost risk *1,*2,*3

Input (low–high)	KEPCO CfD	Direct PPA	Merchant
CAPEX (KRW k/kW) 1,396–1,892	23.7% ~ 8.8%	23.3% ~ 10.2%	7.3% ~ 2.7%
Capacity factor 14%–18%	6.6% ~ 17.9%	8.3% ~ 18.2%	2.0% ~ 5.5%
SMP base 90–148	11.9% ~ 12.7%	13.1% ~ 13.6%	0.9% ~ 8.2%
OPEX (% of CAPEX) 1%–2%	14.5% ~ 9.9%	15.2% ~ 11.3%	4.7% ~ 2.7%

Source: *1 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *2 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *3 KSJ Research
Note: One input at a time at its range ends (debt re-sized). CAPEX low = 2022 actual cost, high = matched to the 136 LCOE · CF range is an author sensitivity

How to read it: the KEPCO route earns through leverage, the PPA through price *1

Same plant, different capital structures

- KEPCO: price 147.7 vs LCOE 125.5, 80% debt → 12.2%
- PPA: 13.3% at the corridor mid KRW 156.5/kWh (no termination) — it needs KRW 152.3/kWh to match KEPCO (p5)
- Merchant: lenders give only 38% and SMP is low → 3.7%
- Against the 8% hurdle: KEPCO CfD · Direct PPA clear, Merchant miss — between contracts it is price vs termination risk

Source: *1 KSJ Research

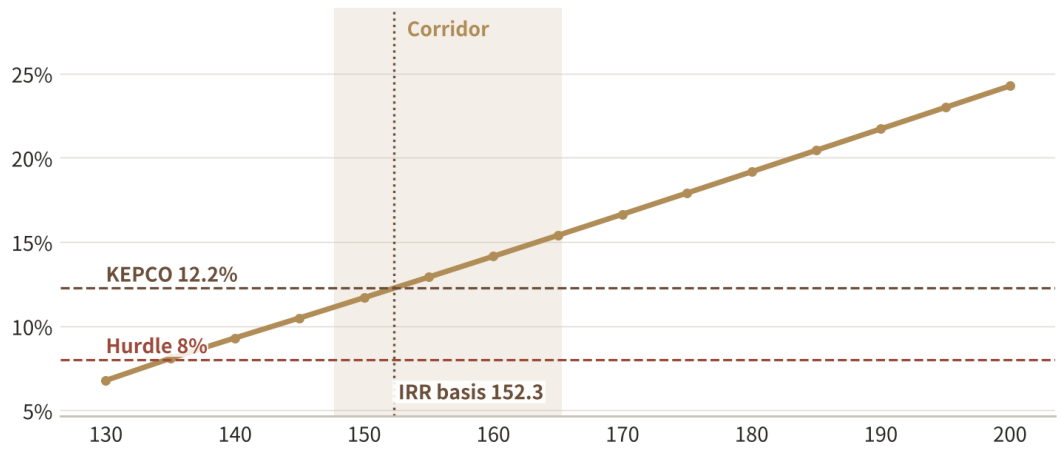
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PPA negotiation: the seller needs +5–16, but the buyer's 6~17 room is below the 40 public example (B)

IRR by price → required price by termination × spread (NPV) → buyer's seasonal saving → what it takes to close

Equity IRR by PPA price: equals KEPCO's IRR at KRW 152.3 — near the corridor floor *1,*2,*3

Unit: Equity IRR (no termination)



Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KSJ Research
Note: Debt re-sized at each price (PPA terms). All 'required prices' here are NPV-based (table) — 149.7 with no termination and no spread. The IRR basis (152.3) is higher because equity sizes differ (gearing 80% vs 75%)

Required premium: +5–16 at 1% termination — the risk spread matters as much as the termination rate *1,*2,*3,*4

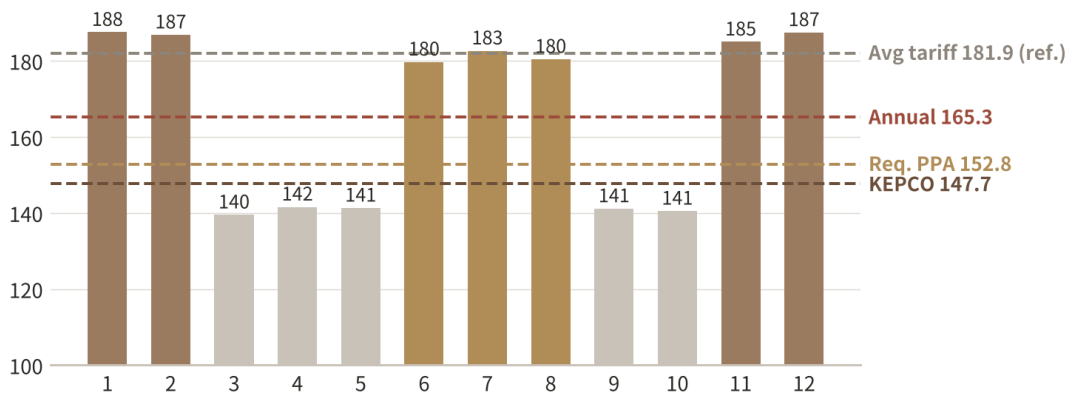
Annual term.	Premium +0pp	Premium +1pp	Premium +2pp	Room for charges
0.0%	+2.0	+7.4	+12.8	9.3~20.1
0.5%	+3.5	+9.1	+14.6	7.5~18.6
1.0%	**+5.1**	**+10.8**	**+16.5**	**5.6~17.0**
2.0%	+8.3	+14.3	+20.3	1.8~13.8
3.0%	+11.6	+18.0	+24.3	−2.2 ~ 10.4
5.0%	+18.8	+25.9	+32.8	−10.7 ~ 3.3

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KEPCO Monthly Statistics no.566, table 16-1 (2025 monthly solar output) *4 KSJ Research
Note: Required price (NPV) = PPA price at which expected equity NPV equals KEPCO's equity NPV at 8%. PPA flows discounted at the hurdle plus a spread (private-counterparty premium, judgement grid). Constant annual termination during the PPA, SMP afterwards, debt kept at no-termination sizing. Premium = required − KEPCO price; room = solar-hour saving 165.3 × (1 + 2.7% fund levy) = 169.8 − required (same basis as levy-inclusive charges)

Seasonal saving vs KEPCO price: spring/autumn (43% of output) at 140.9, below KEPCO's 147.7 *1,*2,*3,*4,*5

spring/autumn 140.9 summer 181.0 winter 186.6

Unit: KRW/kWh (monthly saving = energy rate + fuel adjustment + climate charge)



Source: *1 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *2 KEPCO Monthly Statistics no.566, table 16-1 (2025 monthly solar output) *3 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *4 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *5 KSJ Research
Note: Post-reform industrial (B) HV-A option II energy rates × daily output shape (judgement) × 2025 calendar (Sundays off-peak all day, 50% off 11:00–14:00 on spring/autumn weekends); seasonal and annual values weighted by 2025 monthly output. Required PPA at 1% termination, no spread. If the discount ends after 2030: 168.1 a year · Provisional: Daily output shape exponent, Saturday time-band rule, Sunday/holiday rate rule, Fuel cost adjustment (Q4 2026) pending official source (A3)

The negotiation: the price can be agreed (room 6~17) but fails once network charges (~40 public example, B) are paid — it turns on the buyer's RE100 willingness to pay *1,*2,*3

What the IC will ask

- A 17.6 corridor vs the seller's +5–16 → buyer room 5.6~17.0 (40 public example)
- To close, the buyer must pay 0–9 (KPX settlement only) to 23–34 (public example) per kWh for RE100 value — the 181.9 average tariff hides this
- Judgement that moves it: PPA gearing 65% → +9, 80% → +3 (1%, no spread)
- In a corporate survey, opaque network-charge setting is the No.2 reason to avoid PPAs (45.2%); KPX has commissioned a direct-PPA policy study (Newsis, 4 May 2026)

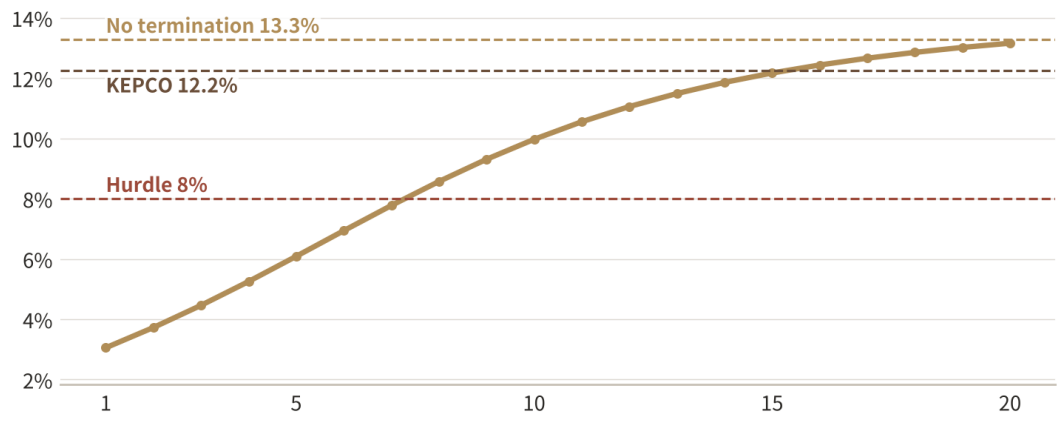
Source: *1 Newsis (4 May 2026): SFOC corporate survey, KPX direct-PPA study *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KSJ Research

Offtaker risk: a PPA shorter than 15 years underperforms KEPCO — termination payments and guarantees must cover early breaks

Equity IRR by termination year → hurdle-holding term → Vol.2 credit → contractual mitigants

Equity IRR by termination year: a PPA that ends before 15 full years does worse than KEPCO *1

Unit: Equity IRR



Source: *1 KSJ Research
Note: x-axis = termination year N (SMP 113.2 from year N = PPA lasted N−1 years). PPA at corridor mid KRW 156.5/kWh, debt fixed (no refinancing), no termination payment

To keep the 8% hurdle the PPA must last 7+ years *1

Termination	Equity IRR	vs KEPCO
Year 1 (PPA lasted 0 yrs)	3.1%	below
Year 3 (PPA lasted 2 yrs)	4.5%	below
Year 5 (PPA lasted 4 yrs)	6.1%	below
Year 8 (PPA lasted 7 yrs)	8.6%	below
Year 10 (PPA lasted 9 yrs)	10.0%	below
Year 15 (PPA lasted 14 yrs)	12.2%	below
Year 20 (PPA lasted 19 yrs)	13.2%	above
No termination	13.3%	—

Source: *1 KSJ Research
Note: After termination the plant earns SMP, so early breaks hurt most — termination payments prop up the left side of this curve

Offtaker = Vol.2's data centre: lender min DSCR 1.76x, power passed to tenants *1

Item	Value
Data-centre SPV min DSCR (Vol.2 Korea)	1.76x
Data-centre equity IRR (Vol.2)	15.5%
Power cost treatment (Vol.2)	Billed to tenants (100%)
PPA volume / data-centre use	15%

Source: *1 KSJ Research Vol.2 (Oct 2026)
Note: The PPA is ultimately paid from tenants' power bills — if leases are shorter than the PPA, the gap becomes termination risk

Mitigants: contract terms that cut termination risk and loss *1

What to hold in the term sheet (A1)

- Termination payment: covers part of the PPA-vs-SMP gap for the remaining term
- Credit support: parent guarantee or LC, or align lease power clauses with the PPA term
- Assignment: the PPA follows the data centre on sale or refinancing
- Change in law: a renegotiation rule if network-charge reform changes the buyer's cost

Source: *1 KSJ Research

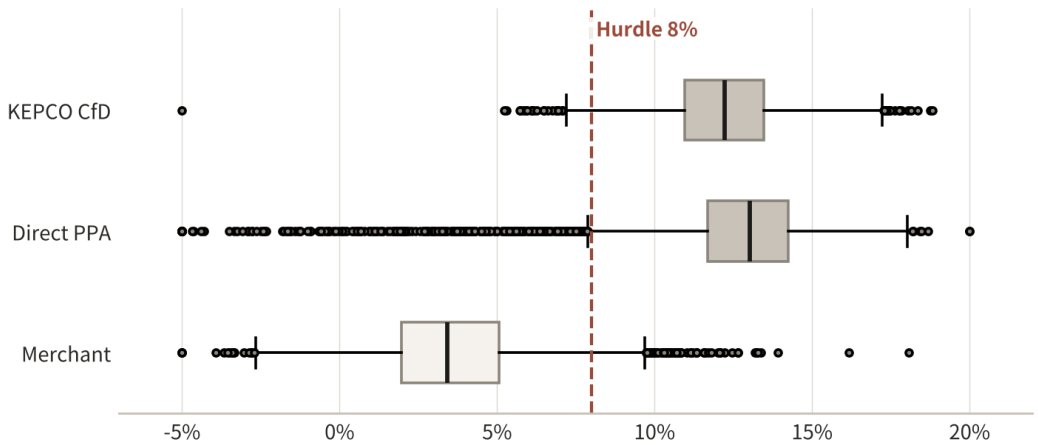
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Monte Carlo: PPA equity IRR P50 13.0% · P5 6.4% — below KEPCO 12% of the time; merchant loses money 5% of the time

5,000 runs × 3 routes · seed 42 · SMP (log AR1) × output × PPA termination

Equity IRR (5,000 runs): the PPA has the higher median but a weaker bottom 5% (P50 13.0% vs 12.2% · P5 6.4% vs 9.1%) *1,*2

IQR box · median Unit: Equity IRR



Source: *1 KPX EPSIS system marginal price (monthly weighted and hourly, mainland) *2 KSJ Research
Note: Common random numbers (SMP, output) across routes — differences come from the contracts only. PPA termination 1% a year. Shown −5% to 20%

Probability below the 8% hurdle: KEPCO 1% · PPA 7% *1

Route	P5	P10	P50	P90	< hurdle	Loss (IRR<0)
KEPCO CfD	9.1%	9.9%	12.2%	14.7%	1%	0%
Direct PPA	6.4%	9.9%	**13.0%**	15.3%	7%	1%
Merchant	0.0%	0.7%	3.4%	6.7%	96%	5%

Source: *1 KSJ Research
Note: The PPA underperforms KEPCO on the same path 12% of the time. Debt fixed at base sizing — equity funds shortfalls (no lock-up or default mechanics in the model)

Inputs: SMP volatility 0.23 and mean reversion 0.72 estimated from 24 years of annual SMP (2002–2025) *1

Input	Value	Grade
SMP volatility	0.23	Computed from official data
SMP mean reversion (AR1)	0.72	Computed from official data
Annual output variation	5%	Author judgement
Resource estimate error	4%	Author judgement
PPA annual termination	1%	Author judgement
Runs · seed	5,000 · 42	—

Source: *1 KSJ Research
Note: SMP: log AR(1), mean-preserving · output: one resource error × annual noise · termination: geometric within the PPA term

How to read it: the PPA's risk is in the tail, not the average *1

What the distribution says

- KEPCO: narrow (P10 9.9%–P90 14.7%) — only output varies
- PPA: above KEPCO down to P10, but P5 6.4% — early terminations (~12%) form the bottom tail
- Merchant: 5% loss probability — uncontracted new solar is not an equity case
- IC view: settle the price (the buyer's RE100 willingness to pay) first, then tail protection (termination payments, guarantees)

Source: *1 KSJ Research

Implications: the buyer's room of at most ~17 falls short of the ~40 public network-charge example (B) — confirm RE100 willingness to pay and actual charges first, then termination protection

Conclusion · IC checklist · limits · Vol.5 preview

Conclusion: on solar-hour rates the direct PPA fails once network charges are paid [*1,*2,*3]

- Price: the seller needs +5.1–16.5 over KEPCO (1% termination, 0–2pp spread) — the buyer saves only +17.6 in solar hours
- Make-or-break: buyer room 5.6~17.0 on the saving plus fund levy — covering ~40 of charges needs 23–34 per kWh of RE100 value from the buyer
- Risk: early termination — termination payments, guarantees and assignment lower the loss and the premium
- Alternatives: KEPCO at 12.2% (below the corridor-mid PPA's 13.3%), merchant at 3.7%

Source: *1 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *2 SFOC, 'Stranded RE100' (Dec 2021), printed p.9 (PDF p.11) table 2 — KEPCO's third-party PPA charge simulator *3 KSJ Research

IC checklist: this note's assumption vs what a real deal must confirm [*1]

Item	This note	Confirm in a deal	p.
Price	Corridor 147.7–165.3	2027 contract-market cap · actual tariff class and option · measured hourly output · climate and fuel add-ons	3, 5
Network charges	Room 6~17 vs 40 example	Current T&D tariffs · settlements · losses · fund levy · buyer's RE100 willingness to pay	3, 5
Offtaker	Termination grid 0–5%	Rating · lease term · guarantees	6
Output	CF 16% (BNEF/KEEI)	P50/P90 yield report · curtailment	4, 7
CAPEX	KRW 61.1bn (from LCOE)	EPC quote · land · grid connection	4
Shape	Pay-as-produced	24/7 firming cost (storage, KEPCO)	—

Source: *1 KSJ Research

Limits of this note [*1]

- CAPEX: 2022 actual installed cost (KRW 1,396k/kW) implies an LCOE of 100, below KEEI's 115–136 — matched to the LCOE to stay conservative
- Buyer ceiling = solar-hour energy rate (HV-A option II) + fuel adjustment (Q4 2026, press report B, changes quarterly): daily shape and Saturday/holiday rules are judgements, climate charge 9.0 (KEPCO ON, effective 2023-01, official), weekend discount ends 2030 (+2.8 after) · no RE100 value or tariff rises
- Plant sized at 35 MW so all output is used — 24/7 shape risk, storage and curtailment left out
- Debt terms, termination rate and risk spread are unsourced judgements — shown as grids and sensitivities

Source: *1 KSJ Research

Next, Vol.5 (finale): an AI energy asset investment map [*1]

Demand, equipment, PF, IPO and power on one page

- Place Vol.1–4 on one risk-return map — which asset fits which investor
- IC memo format: thesis, risks and mitigants, term sheet
- All assumptions and source grades in one table

Source: *1 KSJ Research

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IC memo: equity in a direct-PPA plant — on hold; KEPCO is the base case until the buyer is shown to pay 23+ per kWh for RE100 value (with the ~40 public charge example)

Summary · points · risks and mitigants · term sheet (illustrative)

Deal summary (illustrative — hypothetical asset) *1

Item	Detail
Asset	Hypothetical 35 MW solar, COD 2028
Revenue	20-yr direct PPA with a data centre, pay-as-produced
Uses · sources	KRW 61.1bn = debt KRW 45.8bn (75%) + equity KRW 15.3bn
Equity return	IRR 13.3% (corridor mid) · Monte Carlo P50 13.0%
Pricing rule	At least KRW 152.8/kWh–KRW 164.2/kWh (KEPCO +5–16, 1% termination, 0–2pp spread)

Source: *1 KSJ Research

Three investment points *1

- Price edge: at the corridor mid, +1.0pp IRR over KEPCO
- Demand: AI data centres need RE100 supply — direct routes get scarcer after the 2027 contract-market shift and the REC issuance deadline (31 Dec 2026)
- Downside: terminated in year 1 and selling at SMP 113, min DSCR 1.06x (debt paid, lock-up territory); at SMP 90 it falls to 0.79x — termination payments and guarantees are essential

Source: *1 KSJ Research

Risks and mitigants *1

Risk	Mitigant
Offtaker termination	Termination payment · parent guarantee/LC · assignment
Charges exceed the buyer's room	Fix the buyer's all-in price with current T&D tariffs and settlements before signing; change-in-law reopener
Output shortfall	P90 debt sizing · curtailment compensation
SMP fall (after termination)	Re-contracting rights · re-bidding into the KEPCO market
Policy change	Pre-act licence + recognition by notice keeps RECs (addendum art. 3(3)) — not this plant (licensed after 2027 by assumption); check licence date and the notice

Source: *1 KSJ Research

PPA term sheet (illustrative) *1

Clause	Proposal
Term	20 yrs (aligned with leases)
Price	At least KRW 156.5/kWh fixed (1% break-even 152.8)
Volume	All output (pay-as-produced)
Termination	A share of the PV of (PPA − SMP) over the remaining term
Credit support	Parent guarantee or a one-year LC
Attributes	Renewable certificates to the buyer — their value must bridge the gap to the buyer's saving

Source: *1 KSJ Research
Note: Illustrative terms, not an actual contract

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Inputs and source grades (1/2): every number carries a grade

Value · range · unit · grade · source (links in the repository input file)

60 inputs: official 20 · computed from official 3 · press report 5 · secondary 0 · judgement 32 *1,*2,*3,*4,*5,*6

Item	Value (range)	Unit	Grade	Source / basis
Capacity	35	MW	**Author judgement**	Author's design: hypothetical 35 MW (AC) solar plant, sized so peak output stays within th…
COD year	2028	year	**Author judgement**	Author's design: development from 2026 → electricity business licence after 1 Jan 2027 (ou…
Installed cost per kW (2022 actual)	1,396 (1,396–1,635)	KRW k/kW	**Official source**	KEEI Basic Research 25-04 (Kong & Jang), PDF p.93: Korean solar installed cost KRW 1,396k/…
Capacity factor	0.16 (0.14–0.18)	ratio	**Official source**	KEEI Basic Research 25-04, PDF p.10 summary and p.31 table 2-1: Korean utility-scale solar…
Annual degradation	0.005 (0.003–0.008)	ratio/yr	**Official source**	NREL, Jordan & Kurtz, Photovoltaic Degradation Rates — An Analytical Review (2012): median…
Solar LCOE	125.5 (115–136)	KRW/kWh	**Official source**	KEEI Basic Research 25-04, PDF pp.9 and 18: Korean solar LCOE KRW 115–136/kWh in 2024 (mid…
WACC (LCOE cross-check)	0.06	ratio/yr	**Official source**	KEEI Basic Research 25-04, PDF p.10 and p.31 table 2-1: Korean solar WACC (estimate) 6% — …
OPEX (% of CAPEX)	0.015 (0.01–0.02)	ratio of CAPEX/yr	**Author judgement**	Author's judgement: O&M plus insurance at 1.5% of CAPEX a year. No OPEX figure in the KEEI…
OPEX escalation	0.02	ratio/yr	**Author judgement**	Author's judgement: inflation
Operating life	25	years	**Author judgement**	Author's judgement: in line with 25-year module warranties
Depreciation life	20	years	**Author judgement**	Author's judgement: straight-line 20 years (statutory useful life to be confirmed)
Corporate tax brackets	2:10,200:20,3000:22,inf:25	KRW 100mn : %	**Official source**	Corporate Tax Act art. 55(1)1 — law.go.kr original (effective 1 Jul 2026, Act No. 21217, a…
Local income tax	0.1	ratio of national tax	Press report / secondary report	Local Tax Act — corporate local income tax at 10% of corporate tax; article original not y…
KEPCO contract price (2026 round-1 cap)	147.7	KRW/kWh	**Author judgement**	Author's judgement (proxy): the 2026 round-1 solar auction cap of KRW 147.686/kWh in the m…
Contract term	20	years	Press report / secondary report	Contract-market reform (promulgated 15 Sep 2026, effective 1 Jan 2027): long-term fixed-pr…
KEPCO industrial average tariff (2025, reference)	181.9	KRW/kWh	**Official source**	KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2: 2025 industrial sales reve…
Industrial (B) time-of-use rates	Industrial (B), post-reform	KRW/kWh	**Official source**	Ministry and KEPCO release 'Seasonal and time-of-use tariff reform' (13 Mar 2026), PDF p.8…
Buyer's tariff class	HV-A option II	tariff class	**Author judgement**	Author's judgement: industrial class as in Vol.2 (average load ~36 MW, industrial (B) abov…
Spring/autumn weekend midday discount	0.5	discount	**Official source**	Same release, PDF pp.3 and 7, annex 1: 50% off energy rates 11:00–14:00 on spring (Mar–May…
Sunday/holiday rate rule	Sunday off-peak all day	rule	**Author judgement**	PDF p.2 chart (HV-A option 2, spring/autumn example): Sundays and holidays at off-peak rat…
Saturday time-band rule	Same as weekdays	rule	**Author judgement**	Author's judgement: the release gives no Saturday time-band rule, so weekday bands are use…
Monthly solar output weights	2025	year	Computed from official data	KEPCO Monthly Statistics no.566, sheet 16-1 (solar output estimates): 2025 monthly solar o…
Daily output shape exponent	1 (0.5–2)	exponent	**Author judgement**	Author's judgement (standard shape): no official hourly solar output in the repository, so…
Fuel cost adjustment (Q4 2026)	5	KRW/kWh	Press report / secondary report	Fuel cost adjustment +KRW 5.0/kWh — ZDNet Korea (21 Sep 2026, citing KEPCO's Q4 2026 rate)…
Fuel adjustment annual cap (KEPCO rule)	5 (-5–5)	KRW/kWh	**Official source**	KEPCO ON 'Cost-linked tariff (fuel cost adjustment)', PRM107D00 capture PDF p.2: the adjus…
Climate & environment charge	9	KRW/kWh	**Official source**	KEPCO ON 'Cost-linked tariff (climate and environment charge)' capture PDF p.2, table 'rat…
Ancillary charges, public example (KEPCO-brokered 3rd-party PPA, solar Case 1)	40	KRW/kWh	Press report / secondary report	SFOC, 'Stranded RE100' (Dec 2021), printed p.9 (PDF p.11) table 2: KEPCO Energy Marketplac…
Power-industry fund levy rate	0.027	ratio of power bill	**Official source**	Electricity Business Act decree art. 36: fund levy = 27/1,000 of the power bill (amended 4…
Direct-PPA ancillary settlement (Jul 2023–Jun 2024)	9.73	KRW/kWh	**Official source**	KPX website Q&A (19 Dec 2023): direct-PPA ancillary settlement rate KRW 9.73/kWh (Jul 2023…
SMP base	113.2 (90.4–148.4)	KRW/kWh	**Official source**	KPX EPSIS SMP (weighted, monthly, mainland) workbook: Sep 2025–Aug 2026 mainland average K…

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *4 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *5 Talen Energy Form 8-K (11 Jun 2025, SEC EDGAR) *6 KSJ Research
Note: Grades: official (gold) = government, public body, statute or SEC original · press report = media citing an official release · secondary (red) = replace before final · judgement (brown) = stated basis, stress-tested

Inputs and source grades (2/2): every number carries a grade

Value · range · unit · grade · source (links in the repository input file)

60 inputs: official 20 · computed from official 3 · press report 5 · secondary 0 · judgement 32 *1,*2,*3,*4,*5,*6

Item	Value (range)	Unit	Grade	Source / basis
SMP trend	0	ratio/yr	**Author judgement**	Author's judgement: nominal SMP trend 0% (shown as a sensitivity)
SMP volatility	0.229	log s.d. (annual)	Computed from official data	KPX EPSIS SMP workbook: 2002–2025 annual SMP (24 years), residual s.d. of a log AR(1) regr…
SMP mean reversion (AR1)	0.718	AR(1) coefficient	Computed from official data	KPX EPSIS SMP workbook: AR(1) coefficient of the same regression 0.718 — mean reversion th…
REC revenue	0	yes/no	**Official source**	Renewable Energy Act amendment (Act No. 21943, promulgated 15 Sep 2026, effective 1 Jan 20…
REC transitional rule for existing holders	1	yes/no	**Official source**	Renewable Energy Act amendment (Act No. 21943), addendum art. 3(2): holders issued RECs by…
Annual output variation	0.05	ratio	**Author judgement**	Author's judgement: year-to-year irradiance variation (independent each year)
Resource estimate error	0.04	ratio	**Author judgement**	Author's judgement: long-run yield estimate error (P50 error)
KTB 3-yr	3.84	%	**Official source**	Bank of Korea ECOS, KTB 3-yr (as-of date)
Credit margin	150	bp	**Author judgement**	Author's judgement: same credit margin as the Vol.2 Korean case
Target DSCR · CfD	1.2	x	**Author judgement**	Author's judgement: government contract (KEPCO counterparty) revenue — low end of contract…
Target DSCR · PPA	1.3	x	**Author judgement**	Author's judgement: raised for the private offtaker's (data-centre SPV) credit risk
Target DSCR · merchant	1.6	x	**Author judgement**	Author's judgement: merchant revenue — reflects price risk
Max gearing · CfD	0.8	ratio	**Author judgement**	Author's judgement
Max gearing · PPA	0.75	ratio	**Author judgement**	Author's judgement
Max gearing · merchant	0.55	ratio	**Author judgement**	Author's judgement
Tenor · CfD	18	years	**Author judgement**	Author's judgement: 2-year tail inside the 20-year contract
Tenor · PPA	18	years	**Author judgement**	Author's judgement: 2-year tail inside the 20-year PPA
Tenor · merchant	12	years	**Author judgement**	Author's judgement
PPA term	20	years	**Author judgement**	Author's judgement: 20 years — overseas hyperscaler nuclear PPAs of 20 years (Meta–Constel…
PPA escalation	0	ratio/yr	**Author judgement**	Author's judgement: fixed price (shown as a sensitivity)
Equity hurdle	0.08	ratio/yr	**Author judgement**	Author's judgement: equity required return (for exit-value discounting and loss probability)
Exit year	10	years	**Author judgement**	Author's judgement: infrastructure fund holding period
Buyer's required return	0.07	ratio/yr	**Author judgement**	Author's judgement: operating-asset buyer's required equity return (exit price)
Data-centre IT capacity	40	MW	**Author judgement**	KSJ Research Vol.2 Korean case design (phase 1 20 MW + phase 2 option 20 MW; author's desi…
PUE	1.3	x	**Author judgement**	KSJ Research Vol.2 assumption (author's judgement)
IT load factor	0.7	ratio	**Author judgement**	KSJ Research Vol.2 assumption (author's judgement)
Talen–AWS PPA notional revenue	18	USD bn	**Official source**	Talen Energy 8-K investor materials (11 Jun 2025), pp.3 and 7: 17-year AWS PPA, ~$18bn not…
Talen–AWS term	17	years	**Official source**	Talen Energy 8-K investor materials (11 Jun 2025), p.3: 17-year contract
Talen–AWS contract capacity	1,920	MW	**Official source**	Talen Energy 8-K press release (11 Jun 2025): 1,920 MW at full volume, to 2042, ramping up…
LevelTen N. America solar PPA	61.4	USD/MWh	Press report / secondary report	LevelTen Energy Q2 2026 North American solar PPA market average $61.40/MWh — TaiyangNews (…

Source: *1 Ministry of Climate, Energy and Environment release (16 Jul 2026, round-1 solar auction) *2 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *3 KEPCO Monthly Statistics no.566 (Dec 2025), tables 18 and 18-2 *4 KEEI Basic Research 25-04 (2025) — LCOE (citing Lee & Lim 2024), CF and WACC (citing BNEF), unit cost *5 Talen Energy Form 8-K (11 Jun 2025, SEC EDGAR) *6 KSJ Research
Note: Grades: official (gold) = government, public body, statute or SEC original · press report = media citing an official release · secondary (red) = replace before final · judgement (brown) = stated basis, stress-tested

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Data notes: official sources first; everything else flagged for replacement

As of 2026-10-20 · KTB 3-yr 3.84% (2026-10-20) · annual nominal model, KRW 100mn, KRW/kWh

8 inputs awaiting official sources — all replaced before final [*1]

Item	Current grade	Official source to use
Local income tax	Press report / secondary report	Local Tax Act — corporate local income tax rate article (law.go.kr original)
Contract term	Press report / secondary report	Ministry of Climate, Energy and Environment release original (15 Sep 2026): contract term
Sunday/holiday rate rule	Author judgement	KEPCO basic supply terms enforcement rules (Sunday/holiday time bands) · 2025 public holiday list (Ministry of the Interior and Safety)
Saturday time-band rule	Author judgement	KEPCO basic supply terms enforcement rules — Saturday peak time bands
Daily output shape exponent	Author judgement	data.go.kr 'KPX hourly solar generation' — replace the standard shape with measured output
Fuel cost adjustment (Q4 2026)	Press report / secondary report	KEPCO Q4 2026 fuel cost adjustment rate, official notice (KEPCO ON). Not a direct-PPA payment item — direct-PPA notice art. 25(3) (A)
Ancillary charges, public example (KEPCO-brokered 3rd-party PPA, solar Case 1)	Press report / secondary report	Direct-PPA notice (MCEE notice 2025-61) art. 25(3)6 'other costs designated by the Minister' — whether any is designated now · update direct-PPA charges with current T&D tariffs and settlement rates
LevelTen N. America solar PPA	Press report / secondary report	LevelTen original press release

Built with an AI coding tool (Claude Code) for cash flows, debt sizing, break-evens, Monte Carlo and report generation; question design, source grading, assumption judgments and number checks by the author.

Source: *1 KSJ Research
Note: Rule: official originals first (government, public bodies, statutes, SEC) · media only when citing an official release · private data provisional only. Code and links: GitHub https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

Buyer ceiling, 4 stages: energy rate 151.3 → +fuel adj. 156.3 → +climate 165.3 → +fund levy 169.8 · sensitivities 162.2–176.0, led by tariff class [*1,*2,*3,*4,*5,*6,*7,*8]

Assumption	Annual saving (KRW/kWh)	Room (vs required 152.8)
① Energy rate only	151.3	−1.4
② + fuel adjustment (press B)	156.3	+3.6
③ + climate (official) = price cap	165.3	+12.6
④ + fund levy (× 1.027) = room basis	169.8	+17.0
Daily shape p = 0.5	166.0	+17.7
Daily shape p = 2	163.8	+15.5
Class: HV-A option I	170.7	+22.5
Class: HV-A option III	162.2	+13.8
Class: HV-B option II	176.0	+28.0
Saturday peak → mid-peak	163.8	+15.4
Spring/autumn weekend discount ends (2031+)	168.1	+19.9
Ref.: pre-reform rates (to 15 Apr 2026)	181.0	+33.1

Source: *1 Ministry of Climate, Energy and Environment and KEPCO release (13 Mar 2026), annex 2: industrial (B) time-of-use rates *2 KEPCO Monthly Statistics no.566, table 16-1 (2025 monthly solar output) *3 KEPCO ON cost-linked tariff (fuel cost adjustment) — annual band on the adjustment rate *4 SFOC, 'Stranded RE100' (Dec 2021), printed p.9 (PDF p.11) table 2 — KEPCO's third-party PPA charge simulator *5 Direct-PPA notice (MCEE 2025-61), art. 25(3), 9(4) *6 Electricity Business Act decree art. 36 *7 Kim & Chang newsletter (13 Sep 2022): direct-PPA notice in force *8 KSJ Research
Note: ①–③ = base assumptions (HV-A option II · shape p = 1 · Saturday as weekday) adding per-kWh charges in turn; ④ = ③ × (1 + 2.7% fund levy). Gold = PPA price ceiling (③) and the room used (④). Sensitivities show the avoided rate with add-ons (14.0); room on the ④ basis (× (1 + 2.7%) − required PPA), one assumption at a time. Required PPA at 1% termination, no spread. The +5 fuel adjustment is the KEPCO rule's annual cap (±5, KEPCO ON, A) — holding it for 20 years puts the buyer ceiling at its highest, and it still does not cover network charges (room at most 17.0 < 40 example). Public example 40 = KEPCO-brokered third-party PPA, mid/large solar Case 1 (B; generator outside the capital region on the HV distribution grid, user in the capital region; charge date not stated (KEPCO simulator, report published Dec 2021; only the 136 generation cost is 2020-based)) — SFOC printed p.9 (PDF p.11), table 2: T&D base 18 · usage 7 · losses 3 · settlement 4 · trading fee 0 · welfare/special discounts 3 · fund levy 6 (table total 40 vs item sum 41, rounding). A direct-PPA buyer pays T&D use charges, trading fee, settlement charges, power-industry fund levy (direct-PPA notice art. 25(3) (MCEE notice 2025-61, A); Kim & Chang 13 Sep 2022 (B) as support; item 6 'other costs designated by the Minister' — designation TODO(data)) — no fuel adjustment or climate charge → the ceiling with the 14.0 added is the like-for-like comparison. The example's welfare/special discounts 3 is not a notice item, so direct-PPA charges may be lower than the example. Proviso: items 2 and 4 drop out if the T&D grid is not used · contract term in 1-year units (art. 9(4)). The 40 example includes a 6 fund levy; KEPCO bills carry it too (2.7%, decree art. 36, A) → room on the avoided rate × (1 + 2.7%) = 169.8 (+4.5, per-kWh approximation); it cancels in the 165.3 price ceiling as the PPA payment bears it too.

Glossary: the language of buying and selling power

Prices KRW/kWh · amounts KRW 100mn · U.S. USD/MWh

Glossary

Term	Definition
PPA	Power purchase agreement — seller and buyer fix price, term and volume directly
CfD	Contract for difference — settles contract vs market price; from 2027, long-term contracts with KEPCO
SMP	System marginal price — Korea's hourly wholesale power price
REC	Renewable energy certificate under RPS. Issuance deadline 31 Dec 2026; REC article lapses end-2029; existing holders and recognised licensees keep the former rules (RE Act addendum arts. 2–3)
RPS	Renewable portfolio standard — from 2027 new plants go through the contract market (art. 12-14)
LCOE	Levelised cost of electricity — PV of lifetime costs ÷ PV of output
Network charges	Transmission and distribution charges a direct-PPA buyer pays KEPCO
Offtaker	The buyer under the contract (here the data-centre SPV)
DSCR	Debt service coverage = CFADS ÷ debt service
Sculpting	Shaping debt service so each year's DSCR equals the target
Equity IRR · MOIC	Return on equity · multiple of invested capital
Termination rate	Annual chance the PPA ends early (shown as a grid)
Price corridor	Range between the seller's (KEPCO contract) and buyer's (solar-hour KEPCO energy rate) outside options
Avoided rate	The KEPCO rate a PPA kWh replaces — time-of-use energy rates in solar hours (output-weighted) plus per-kWh add-ons (fuel adjustment, climate charge); no demand charge
Pay-as-produced	Sell what is produced — no 24/7 obligation
Source grade	Official · computed from official data · press report of official release · secondary · judgement

Note: Shaded = core terms

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- This is a personal portfolio for education and research. It is not investment advice or a view on any asset or security.
- The 35 MW solar asset and the data-centre offtaker are hypothetical; no actual contract prices or terms are shown.
- Views are the author's own and do not represent any current or future employer.
- Every input carries a source grade (official · computed from official data · press report of official release · secondary · judgement); see Appendix A2–A4.
- As of 2026-10-20. Methods and code can be reproduced from the repository.
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https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

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